Construction Chemicals

SUBMITTAL DOCUMENT

O-Zinc I

Zinc-Rich Epoxy Primer
Cathodic Protection for Steel



Cathodic
Galvanic protection

Fast Drying
Recoat 5 – 12 h

Weldable
Fabrication friendly

Versatile
Primer or finish

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

O-Zinc I — Zinc-Rich Epoxy Primer Cathodic Protection for Steel

Project Name		**Document No.**	OMX-MS-OZINCI
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Zinc I	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Zinc I on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers preparation, mixing, application, curing and inspection of O-Zinc I (zinc-rich epoxy primer cathodic protection for steel).

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after Part A and Part B are combined.
Recoat interval	Minimum and maximum time between coats to achieve inter-coat adhesion.
WFT / DFT	Wet film thickness measured during application / dry film thickness after cure.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Kit / set	Pre-weighed Part A and Part B supplied to be mixed completely.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.

Role	Responsibility
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Zinc I	5 kg, 20 kg and 25 kg sets (Part A + Part B), metal pails.

5.1 Key product data

Property	Typical Value
Mixing ratio	As recommended — pre-weighed sets
Colour	Grey
Solids content	57 ± 2 %
Density	1.6 kg/L
Pot life	3 hours
Recoat	5 – 12 hours
Full cure	7 days
Flash point	126 °C

6. Tools & Equipment

- Grinder / shot blaster or blasting unit
- Industrial vacuum
- Slow-speed mixer
- Rollers
- Brushes
- Airless spray where specified
- Wet-film comb
- DFT gauge
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge

- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Chemical-resistant gloves
- Goggles
- Overalls
- Respirator where required

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Remove oil and grease by solvent cleaning.
2. Blast-clean to Sa 2½.
3. Remove dust and abrasive residues; coat before re-rusting.
4. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
5. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.
3. Stir Part A thoroughly to disperse the zinc.
4. Add the complete Part B and mix until uniform.
5. Thin max. 5 % only if required.
6. Mix only the quantity that can be applied within the pot life.

11. Application

1. Apply a stripe coat on edges and welds.
2. Apply by brush, roller or spray at 0.15 – 0.20 kg/m² per 50 µm.
3. Stir frequently during application.
4. Work in manageable bays, keeping a wet edge to avoid lap marks.
5. Check consumption by area and wet film thickness regularly during application and record the results.

12. Curing & Protection

- Pot life 3 hours.
- Recoat after 5 – 12 hours; full cure 7 days.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before application	Prepared, sound and clean as specified	QA/QC
Climate	During work	Steel temperature: min. +5 °C, at least 3 °C above dew point	Site Engineer
Consumption	During application	0.15 – 0.20 kg/m² per 50 µm DFT	QA/QC
Finish	After curing	Uniform, no defects	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)

Activity	Inspection / Test	Frequency	Record
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Pinholes / bubbles	Outgassing from porous substrate; air entrapped in mixing	Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash and abrade; recoat in suitable conditions
Poor adhesion	Contaminated, weak or damp substrate; recoat window exceeded	Remove, re-prepare surface and reapply
Soft spots	Incorrect mixing or ratio	Remove affected area and reapply with full kits
Uneven colour / lap marks	Loss of wet edge; different batches	Abrade and apply a further coat from one batch

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Solvent-based: flammable. Keep away from heat, sparks and open flames; no smoking.
- Wear chemical-resistant gloves, safety goggles and overalls; use a respirator with organic vapour cartridges in enclosed spaces.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation during application and curing.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with the recommended thinner before the material cures.

17. Limitations

- Steel temperature: min. +5 °C, at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate prepared and approved	☐ Yes ☐ No	
Batch numbers recorded	☐ Yes ☐ No	
Material mixed as specified	☐ Yes ☐ No	
Applied at specified consumption	☐ Yes ☐ No	
Curing and protection carried out	☐ Yes ☐ No	
Final inspection passed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

O-Zinc I — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Protective Coating

O-Zinc I

Zinc-Rich Epoxy Primer
Cathodic Protection for Steel



Product Description

O-Zinc I is a two-component, zinc-rich epoxy primer that provides cathodic (galvanic) protection to steel. It is non-toxic, dries quickly, is weldable, and can be used as a primer or as a self-finish. It also improves the protection of exposed reinforcement bars before concrete repair.

Uses

- Roads and bridges.
- Workshops and warehouses.
- Car parks and staircases.
- Steel structures and fabrications.
- Protection of exposed reinforcement bars.

Advantages

- Cathodic protection of steel.
- Fast drying and non-toxic.
- Weldable.
- Primer or self-finish.
- Improves rebar protection in repairs.

Application

1. **Surface Preparation**
   Blast-clean steel to Sa 2½; remove dust.
2. **Mixing**
   Mix Part A thoroughly, add Part B and mix until uniform; thin max. 5 %.
3. **Application**
   Apply by brush, roller or spray at 0.15 – 0.20 kg/m² per 50 µm.

Consumption

0.15 – 0.20 kg/m²
per 50 µm DFT
Keep zinc in suspension by stirring during use.

Packaging

5 kg, 20 kg and 25 kg sets (Part A + Part B), metal pails.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio	As recommended — pre-weighed sets	—
Colour	Grey	—
Solids content	57 ± 2 %	—
Density	1.6 kg/L	—
Pot life	3 hours	—
Recoat	5 – 12 hours	—
Full cure	7 days	—
Flash point	126 °C	—
Thinning	max. 5 %	—
Min. steel temperature	+5 °C	—

Consumption

DFT	Consumption
50 µm	0.15 – 0.20 kg/m²
75 µm	0.23 – 0.30 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Steel temperature: min. +5 °C, at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Steel blast-cleaned to Sa 2½.

Application Notes

- Stir continuously to keep zinc in suspension.
- Thin max. 5 % with the recommended thinner.
- Apply a stripe coat on edges and welds.
- Overcoat with the specified intermediate or top coat within the recoat window.

Health & Safety

- Solvent-based: flammable. Keep away from heat, sparks and open flames; no smoking.
- Wear chemical-resistant gloves, safety goggles and overalls; use a respirator with organic vapour cartridges in enclosed spaces.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation during application and curing.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with the recommended thinner before the material cures.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.